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A molecularly engineered electron-selective self-assembled monolayer enhances quasi-Fermi level splitting in inverted perovskite solar cells

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Maximizing quasi-Fermi level splitting is essential for achieving high photovoltage in perovskite solar cells. Conventional strategies that prioritize rapid charge extraction at the perovskite/fullerene interface, can deplete interfacial carrier populations, limiting quasi-Fermi level splitting and photovoltage. Here we demonstrate that rational interface design benefits for slowed extraction rate, preserving interfacial carriers while minimizing non-radiative recombination. We introduce 3PDPA, a molecularly engineered electron-selective self-assembled monolayer at the perovskite/C_{60} interface. 3PDPA slows electron extraction, anchors undercoordinated Pb^{2+} ions and forms stable six-membered hydrogen-bonded rings with FA^+ cations, delivering robust passivation and excellent chemical stability. 3PDPA further enables π–π interactions with C_{60}, improving interfacial contact and reducing potential fluctuations. Inverted cells incorporating 3PDPA achieve efficiencies of 26.82% for 1.53 eV bandgap cells and 21.2% for 1.77 eV bandgap cells, alongside a T_{90} lifetime of ~1,000 h under International Summit on Organic Photovoltaic Stability (ISOS-L-3) stress conditions.

Metal-halide perovskite solar cells (PSCs) have rapidly established themselves as leading contenders in next-generation photovoltaics, combining exceptional power conversion efficiencies (PCEs) with low-cost fabrication and compositional versatility[1–5]. Despite impressive progress, further performance improvements remain constrained by interfacial losses, particularly those that limit the quasi-Fermi level splitting (QFLS), a fundamental determinant of the open-circuit voltage (V_{oc}) and overall device efficiency[6,7].

Under illumination, QFLS reflects the separation of electron and hole quasi-Fermi levels and sets the theoretical upper limit of V_{oc}. However, in real devices, this splitting is reduced by non-radiative recombination, which predominantly occurs at imperfect interfaces[6,8]. Accordingly, strategies that suppress interfacial recombination and preserve high carrier densities are vital to pushing cell efficiencies towards their radiative limit. Whereas great effort has been devoted to enhancing interfacial charge transport[9,10], relatively less attention has

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been paid to the impact of charge extraction on local carrier populations and thus the achievable QFLS.

Considerable advances have been made in optimizing QFLS at hole-selective contacts, where the advent of self-assembled monolayers (SAMs) has transformed interfacial design[7,9,11–14]. Compared to traditional polymeric hole transport materials, such as poly (4-butylphenyldiphenylamine) (poly-TPD) and poly(bis(4-phenyl)(2,4,6-trimethylphenyl)amine) (PTAA)[15–17], SAMs form ordered dipole arrays that regulate interface carrier distribution and create ultrathin, dense passivating layers to suppress recombination[18–21]. This has enabled devices to approach the radiative limit of V_{oc} by substantially enhancing QFLS at the hole–contact interface.

By contrast, the electron-selective interface, often formed using fullerene derivatives such as C_{60}, remains a prominent source of QFLS losses. A key issue lies in the inherently poor chemical interaction between C_{60} and the perovskite surface. As a non-polar organic semiconductor, C_{60} lacks functional groups to passivate undercoordinated ions at the interface and interacts weakly with the inorganic lattice[8,10]. Recent studies have further revealed that C_{60} can react with iodine species migrating from the perovskite, forming deep-level traps that aggravate recombination losses at the immediate perovskite/C_{60} contact[22]. Whereas efforts have been made to mitigate these effects, such as applying alkylammonium halide passivators or chemically modifying the C_{60} surface[10,23–26], these approaches often introduce new challenges. For instance, alkylammonium salts may produce non-uniform two-dimensional perovskite layers and release mobile iodide ions that migrate towards the electrode, thereby compromising device stability[27–29]. Likewise, surface-functionalized C_{60} materials may tune energy levels but still lack sufficient chemical anchoring to prevent interfacial recombination[30–32].

Equally important, but far less appreciated, is the dynamic interplay between charge extraction and carrier distribution at the interface. The prevailing strategy in PSCs has been to accelerate charge extraction, with the aim of reducing carrier residence times and limiting recombination, an approach that has enabled enhanced efficiencies[9,33–35]. However, this strategy implicitly assumes that recombination dominates performance losses. Once interfacial recombination is strongly suppressed, an additional trade-off emerges: excessively rapid electron transfer into C_{60} can deplete the interfacial carrier population, thereby constraining the achievable QFLS. Overcoming this limitation requires a shift in design principles—towards interfaces that preserve a high interfacial carrier density while simultaneously minimizing non-radiative recombination.

Here we report a molecularly engineered electron-selective SAM, 3PDPA, that enables an interfacial design strategy in which electron extraction is intentionally slowed for PSCs. By simultaneously slowing electron extraction and suppressing interfacial recombination, 3PDPA preserves high interfacial carrier density and enhances QFLS. In addition, its robust chemical bonding and π–π interaction with C_{60} layers minimize interfacial electronic disorder and enhance interfacial stability. Devices incorporating 3PDPA show markedly enhanced V_{oc}, efficiency and long-term operational stability under International Summit on Organic Photovoltaic Stability (ISOS-L-3) stress.

Results and discussion

Molecular design and synthesis

We design and synthesize an electron-selective SAM, designated as 3PDPA, with its chemical structure shown in Fig. 1a. The synthetic route and characterizations are detailed in Supplementary Note 1 and Supplementary Figs. 1–4. The terminal unit incorporates a phthalimide electron acceptor, which enables selective electron extraction while simultaneously offering a planar aromatic scaffold that supports π–π interactions[20,36]. This design is expected to facilitate templated molecular ordering at the interface, promoting uniform deposition of C_{60}. To accommodate the intrinsically rough morphology of the top perovskite surface, we introduce a shortened propyl alkyl spacer, which reduces conformational disorder without compromising dipole integrity. As the anchoring group, we select phosphonic acid due to its well-established capability to chemically bind to the perovskite lattice and effectively passivate interfacial defects[37–39]. In addition to these structural and electronic features, the synthesized 3PDPA molecule exhibits excellent solubility in a broad range of organic solvents (Supplementary Fig. 5), ensuring wide process compatibility and offering an extended fabrication window.

To gain structural insights into the molecular conformation and self-assembly behaviour, we grow single crystals of 3PDPA and resolve their structure via X-ray crystallography (Fig. 1b, Supplementary Fig. 6 and Supplementary Table 1). The crystal structure reveals a dihedral angle of 112.8° between the aromatic core and the alkyl chain. This geometry favours exposure of the π-conjugated surface when the molecule is anchored to the perovskite, thus enabling effective π–π stacking with the subsequently deposited C_{60}. A prominent intermolecular C–H···π interaction of 3.18 Å is observed, supporting the formation of a densely packed, well-ordered SAM. This structural orientation is expected to act as a molecular template that directs the orderly assembly of fullerene molecules[7,40].

To elucidate the surface binding mechanism of 3PDPA on perovskite surface, we perform electrostatic potential (ESP) mapping and charge distribution analyses (Supplementary Figs. 7 and 8), revealing that the P = O moiety bears high electron density and thus a strong coordination affinity towards undercoordinated Pb^{2+} ions. This theoretical prediction is supported experimentally by Fourier transform infrared (FTIR) spectroscopy (Fig. 1c), which shows a redshift in the P = O stretching vibration from 1,286 to 1,291 cm^{-1} after 3PDPA interacts with the perovskite. X-ray photoelectron spectroscopy (XPS) further supports this, showing a downshift in the Pb 4*f* binding energy and an upshift in the P = O peak (Supplementary Fig. 9), consistent with electron donation from the P = O group to Pb^{2+}. A new feature at ~531.7 eV also appears, characteristic of a -P = O···Pb coordination bond[41,42].

In parallel, 3PDPA also interacts with organic cations at the perovskite surface through hydrogen bonding. The asymmetric P–OH stretching mode undergoes a 5 cm^{-1} blueshift in FTIR (Fig. 1c), indicative of hydrogen bonding between the phosphonic acid and the formamidinium (FA^+) cations. Liquid-state proton nuclear magnetic resonance (^1H NMR) (Fig. 1d) further supports this interaction: upon adding 3PDPA, the FAI resonance peak splits at 8.78 ppm and remains unchanged after introducing PbI_2. Solid-state NMR likewise reveals the presence of hydrogen-bonding motifs in the film state (Supplementary Fig. 10). Differential charge density calculations (Supplementary Fig. 11) show that protonated FA^+, acting solely as a hydrogen bond donor, forms a stable six-membered, two-site hydrogen bond with the -P-O group of 3PDPA, altering the local proton environment and highlighting the specificity of this interactions[43,44]. The blueshift of the –C = N stretching mode in the FTIR spectra of FAI (Supplementary Fig. 12) further corroborates this hydrogen-bonding interaction.

Moreover, we investigate the interaction between 3PDPA and C_{60}. The FTIR spectrum of C_{60} (Fig. 1e) shows a shift in the characteristic F_{1u} mode from 1,181 to 1,183 cm^{-1} upon interaction with 3PDPA. This subtle blueshift is attributed to π–π stacking interactions[45], whereby electronic coupling between the conjugated core of 3PDPA and C_{60} modulates the vibrational response of the fullerene, reflecting enhanced molecular interaction at the interface.

Taken together, these results demonstrate that 3PDPA robustly anchors onto the perovskite surface through a combination of coordination bonding with Pb^{2+} and hydrogen bonding with FA^+ (Fig. 1f). This dual interaction not only passivates surface defects but also establishes a structurally coherent molecular interface, providing an ideal template for the orderly deposition of subsequent C_{60} layers. Notably, under spin-coating conditions, 3PDPA is unlikely to form a perfect monolayer but rather a partially ordered, multi-molecular assembly—a

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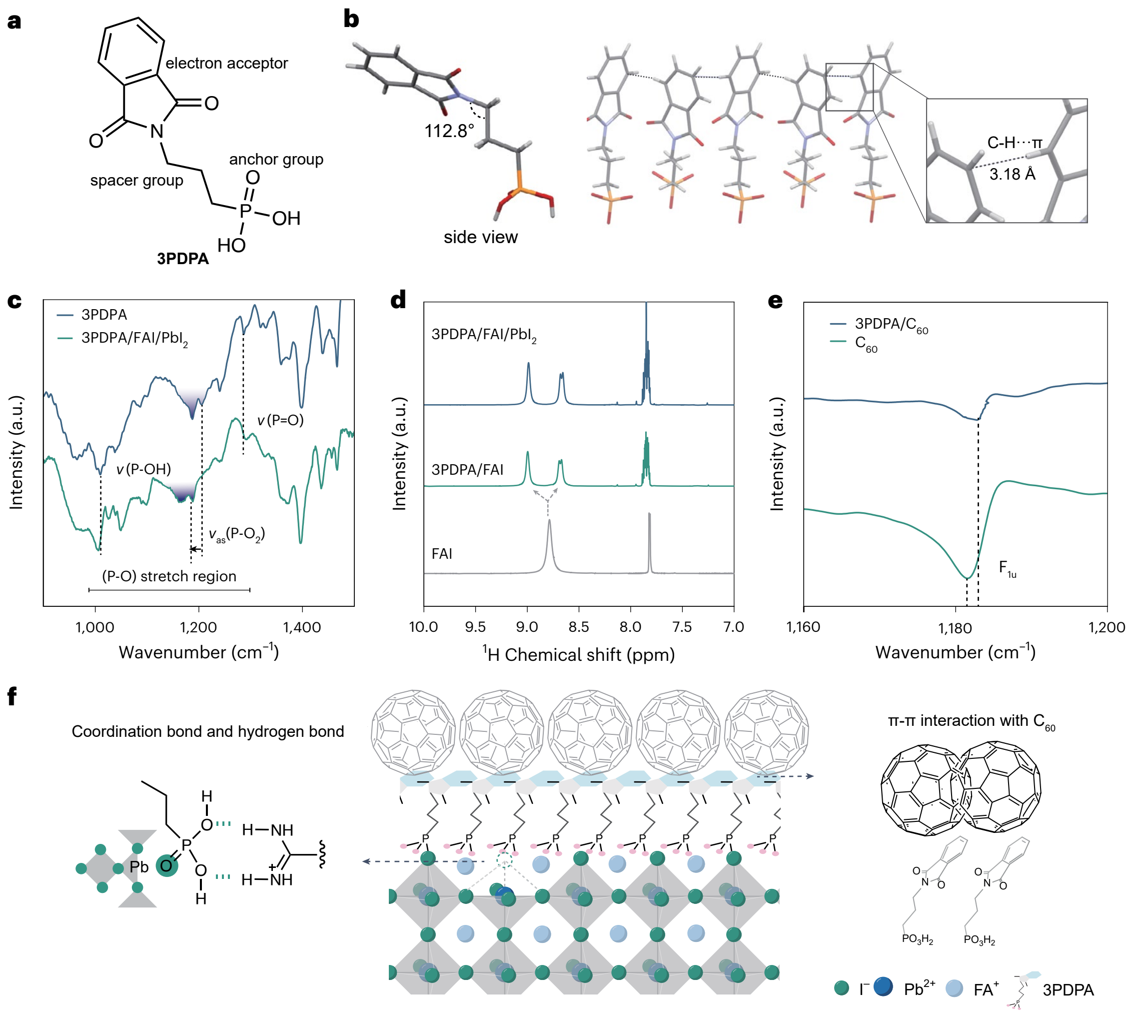

Fig. 1 | Molecular design and interfacial interactions. **a**, Schematic diagram of the molecular structure of 3PDPA. **b**, Side view and single crystal structure of the 3PDPA molecule (enlarged view showing details of π–π stacking). **c**, FTIR spectra of 3PDPA and 3PDPA/FAI/PbI_2 films on glass substrates. (the blue shading indicates that both peaks correspond to -P-O_2). **d**, Liquid ^{1}H nuclear magnetic resonance spectra of FAI, 3PDPA with FAI, 3PDPA with FAI and PbI_2 dissolved in DMSO-d_6. **e**, FTIR spectra of C_{60} powder and C_{60} powder with 3PDPA after grinding, F_{1u} is one of the four classic infrared-active (IR-active) vibrational modes of C_{60}. **f**, Schematic illustration of the interaction of 3PDPA with perovskite and C_{60}. The left inset highlights the 3PDPA enabled iodide defect passivation and the formation of a six-membered hydrogen-bonded ring with FA^+. The right inset represents the π-π interaction of 3PDPA with C_{60}. Panel **b** generated with Mercury[55].

feature that may enhance surface coverage and contribute to improved interfacial quality[46].

Mitigating surface potential fluctuations at the perovskite/C_{60} interface

Surface potential inhomogeneity at the perovskite/fullerene interface poses a major challenge for efficient and uniform carrier extraction in PSCs[47]. Such inhomogeneity arises primarily from poor chemical interaction and weak electronic coupling between C_{60} and the perovskite surface, which can lead to energetic disorder, trap-assisted recombination and ultimately reduced device performance[8]. To investigate whether 3PDPA can mitigate these interfacial inhomogeneities, we employ Kelvin probe force microscopy (KPFM) to map the surface potential evolution through successive interface configurations. The pristine perovskite film exhibits pronounced spatial fluctuations in contact potential difference (CPD), with a variation of ~36 mV (Fig. 2a,e), indicative of surface trap states and compositional inhomogeneities. Upon direct deposition of C_{60}, the surface retains similar topography (Fig. 2b) and the CPD variation across the film remains high (~ 40 meV; Fig. 2f). This suggests that C_{60} forms an electronically disordered interface, probably due to its non-polar character and lack of specific anchoring sites on the perovskite surface.

Introducing a 3PDPA interlayer markedly improves the interfacial landscape. Upon deposition onto the perovskite surface, 3PDPA does not induce noticeable change in morphology (Fig. 2c) but substantially reduces CPD variation (Fig. 2g), indicating improved electrostatic homogeneity and effective passivation of surface defects. Subsequent deposition of C_{60} films onto the 3PDPA-treated substrates exhibits larger domain sizes, lower roughness and improved morphological uniformity (Fig. 2d and Supplementary Fig. 13), which we attribute to π–π interaction-enhanced growth of C_{60}. Crucially, the C_{60} film grown on the 3PDPA-treated surface yields a highly uniform CPD distribution (~ 10 mV; Fig. 2h), in stark contrast to the broader variations observed in the absence of 3PDPA. This confirms that 3PDPA not only stabilizes the underlying perovskite surface but also facilitates a more energetically ordered C_{60} interface. For comparison, perovskite films treated with representative ammonium salts such as PEAI and propane-1,3-diammonium iodide ($PDAI_2$) show minimal

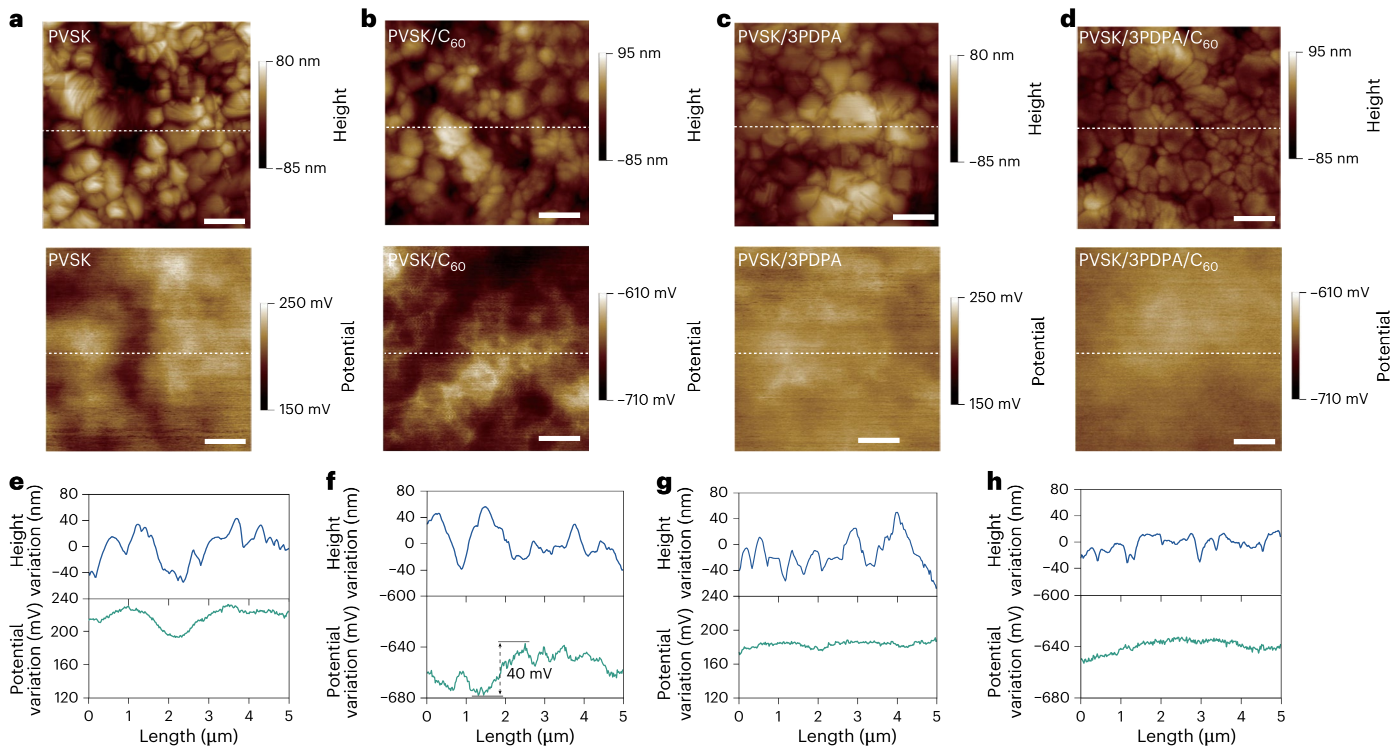

Fig. 2 | Surface morphology and potential distribution. a–d, Topography and surface potential maps of perovskite (PVSK) (**a**), perovskite/C_{60} (**b**), perovskite/3PDPA (**c**) and perovskite/3PDPA/C_{60} (**d**) films on FTO substrates. Scale bars, 1 μm. **e–h**, Height and potential line profiles corresponding to the white dashed lines shown in **a–d**, respectively.

changes in morphology yet retain substantial CPD variability and fail to regulate C_{60} growth, leading to pronounced CPD non-uniformity after fullerene deposition (Supplementary Fig. 14). These results underscore that 3PDPA establishes a distinct interfacial environment that conventional ammonium-halide-based modifiers do not achieve. We further investigate the electronic structure of the perovskite films using photoemission yield spectroscopy in air and Kelvin probe measurements (Supplementary Fig. 15). Incorporation of 3PDPA causes modest changes in the absolute positions of the valence and conduction band edges, whereas the Fermi level shifts by approximately 80 meV towards the vacuum level. This non-rigid shift suggests that the effect arises mainly from defect passivation and suppressed interfacial recombination induced by 3PDPA, rather than from a pure dipole-induced realignment.

Balancing interfacial carrier population and extraction to optimize QFLS

Having demonstrated that 3PDPA establishes a chemically robust and electrostatically uniform interface, we now investigate how this molecular interlayer regulates interfacial carrier populations and its impact on QFLS. Time-resolved photoluminescence (TRPL) measurements reveal marked differences in carrier lifetimes depending on the interface design (Fig. 3a). The pristine perovskite film exhibits a long mono-exponential decay (~2.2 μs), indicative of the high quality of the perovskite film. Deposition of C_{60} directly onto perovskite dramatically shortens the TRPL lifetime to 37 ns, suggesting the strong electron-accepting nature of C_{60} (ref. 36). In contrast, 3PDPA alone quenches the TRPL to ~1 μs, demonstrating its role as an electron-accepting SAM with much weaker extraction than C_{60}. When 3PDPA is inserted between perovskite and C_{60}, the decay lifetime extends to ~116 ns, indicating that 3PDPA slows interfacial charge extraction while still facilitating electron transfer. This is further supported by steady-state photoluminescence spectra (Supplementary Fig. 16).

The ultra-fast charge transfer between perovskite and C_{60} is further confirmed by transient absorption (TA) spectroscopy (Fig. 3b,c). It is worth to note that the samples are pumped with 500 nm on the perovskite side, which is on the opposite of the perovskite/C_{60} interfaces. Interestingly, the photogenerated carrier induced signal is reduced, along with a fast decay and a slightly blueshift of the TA peak, which is a sign of ultra-fast charge transport within perovskite layers and the transfer between perovskite and C_{60} (Supplementary Fig. 17). As a control, the TA decay of a perovskite-only film without C_{60} exhibits much slower recovery (Supplementary Fig. 18), confirming that the rapid early-time bleach recovery in the perovskite/C_{60} and perovskite/3PDPA/C_{60} stacks arises from interfacial electron transfer rather than intrinsic carrier relaxation. By contrast, the insertion of 3PDPA reduces the charge transfer rate at the interface, indicating the effective modulation of charge transfer between perovskite and C_{60} with negligible charge-carrier losses. To distinguish whether this slower decay originates from retarded charge extraction or suppressed recombination, we perform fluence-dependent TA measurements (Supplementary Fig. 19). We find that under high excitation fluence, where trap-assisted recombination is largely suppressed, 3PDPA-modified samples still exhibit slower decay, confirming that 3PDPA intrinsically reduces interfacial charge extraction rate. We quantify the QFLS by measuring photoluminescence quantum yield (PLQY) of the different film structures (Supplementary Fig. 20). The introduction of 3PDPA markedly enhances the PLQY, corresponding to a 30-meV reduction in QFLS loss. This improvement occurs even at sub-monolayer C_{60} coverages (Supplementary Fig. 21), where long-range ordering is absent, indicating that the enhanced QFLS primarily originates from the chemically defined perovskite/3PDPA interface itself rather than from improved C_{60} film ordering, consistent with the finding that the most prominent recombination occurs at the immediate perovskite/fullerene contact[8].

To understand these observations, we conduct device simulations using a full device FTO/hole-transporting layer/perovskite/C_{60}/

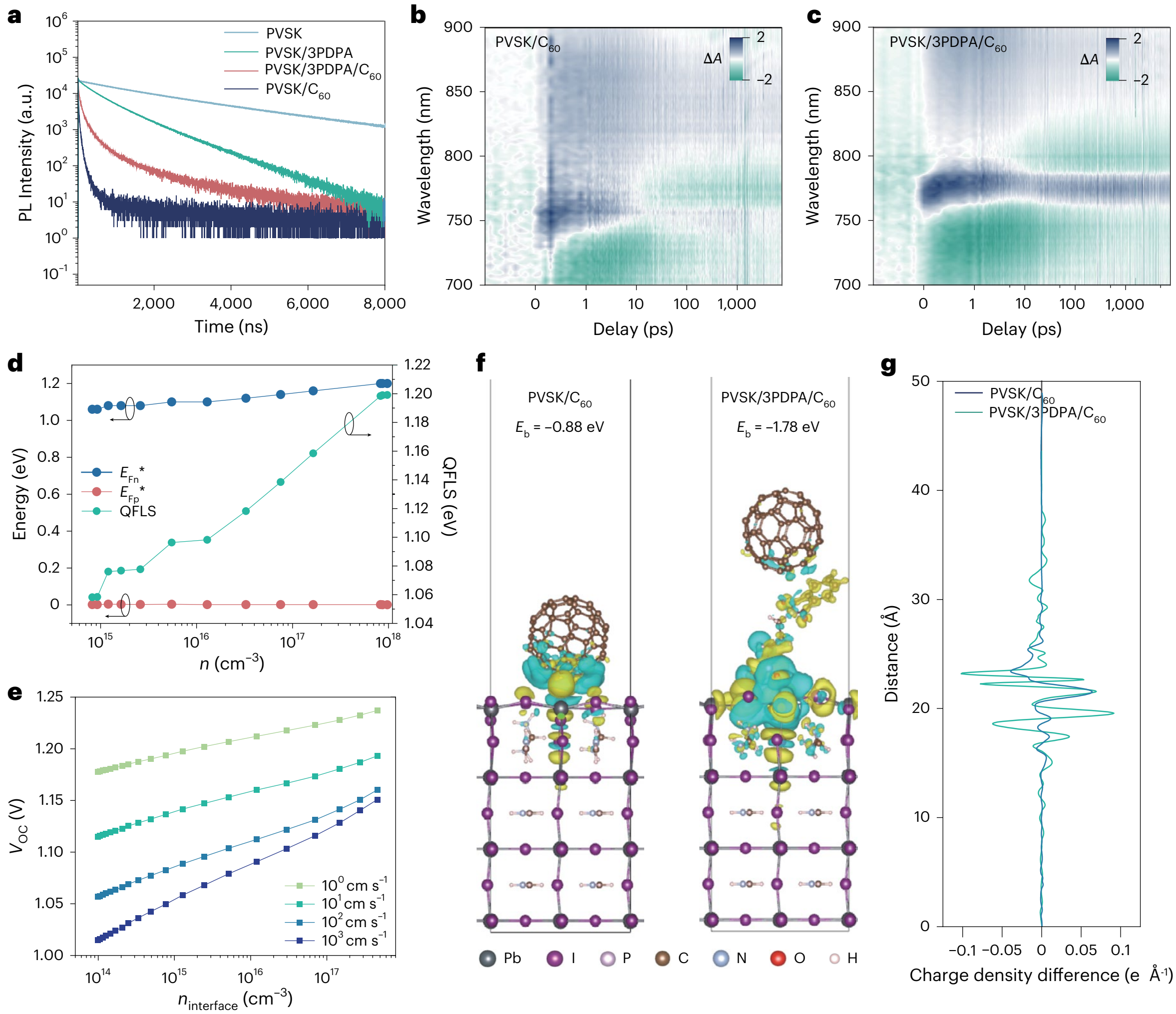

Fig. 3 | Interfacial carrier dynamics and their impact on QFLS. a, TRPL spectra of neat perovskite (PVSK), perovskite/3PDPA, perovskite/C_{60} and perovskite/3PDPA/C_{60} films on glass substrates. **b**,**c**, Femtosecond TA spectra of perovskite/C_{60} (**b**) and perovskite/3PDPA/C_{60} (**c**) films on glass substrates. **d**, Simulated evolution of quasi-Fermi level positions and corresponding QFLS as a function of n. **e**, Plots of V_{oc} versus n under different recombination rates. More simulation details are described in Methods. **f**, Charge density difference distributions of perovskite (with PbI_2-terminated $FAPbI_3$ surface)/fullerene slabs, green areas indicate electron dissipation, while yellow areas represent electron accumulation. **g**, Corresponding charge density difference profiles along the z axis. Panel **f** created with VESTA 3[59].

Cu stack, introducing a thin sublayer at the perovskite/C_{60} interface to emulate varying carrier densities. The simulation details are given in Methods. By logarithmically varying the doping concentration of this sublayer from 1×10^{16} to 1×10^{18} cm^{-3}, we extract the corresponding QFLS and local electron and hole quasi-Fermi levels (E_{Fn}^* and E_{Fp}^*) for each condition (Fig. 3d). We observe the QFLS increases monotonically with interfacial carrier density (n), driven by a progressive elevation of $E_{F_n}^*$. This highlights the central importance of preserving interfacial carrier populations to maximize QFLS.

To further elucidate the role of interfacial recombination, we introduce a range of surface recombination velocities ($S = 1-10^3$ cm s^{-1}) at the perovskite/C_{60} interface in our simulations, mimicking different levels of interfacial defect passivation (Fig. 3e). Across all scenarios, we observe that increasing the n leads to a corresponding rise in the V_{oc}, underscoring the importance of carrier accumulation at the interface. Importantly, the effectiveness of this accumulation is strongly modulated by the degree of recombination. At higher recombination velocities, the V_{oc} becomes increasingly sensitive to variations in n, but also more easily limited by trap-assisted recombination. Conversely, at lower recombination velocities, representing well-passivated interfaces, V_{oc} continues to increase steadily with n, with fewer losses to non-radiative pathways. These results highlight the dual role of 3PDPA in boosting QFLS and V_{oc}: first by passivating interfacial defects and thereby reducing recombination velocity and second by slowing the rate of electron extraction into C_{60}, which helps preserve a higher interfacial carrier population. Together, these effects show that high QFLS (and thus V_{oc}) relies on simultaneously slowing charge extraction to avoid carrier depletion and suppressing defect-mediated losses. This analysis reconciles seemingly divergent literature reports by showing that fast extraction is beneficial in recombination-limited regimes, whereas slowed extraction becomes advantageous once recombination velocities are sufficiently low. Whereas the precise quantitative threshold for optimal extraction requires further study, the combined effects of slowed charge transfer and effective defect passivation establish a broadly applicable interfacial design principle.

To understand the underlying mechanism by which 3PDPA modulates interfacial charge transfer, we perform density functional theory (DFT) calculations to investigate the interfacial interactions involving 3PDPA, perovskite and C_{60} (Fig. 3f). Owing to the non-polar nature of C_{60}, its interaction with the perovskite is comparatively weak, exhibiting restricted electronic coupling and yielding a calculated binding energy of 0.88 eV. By contrast, the 3PDPA molecule, featuring a dipole moment

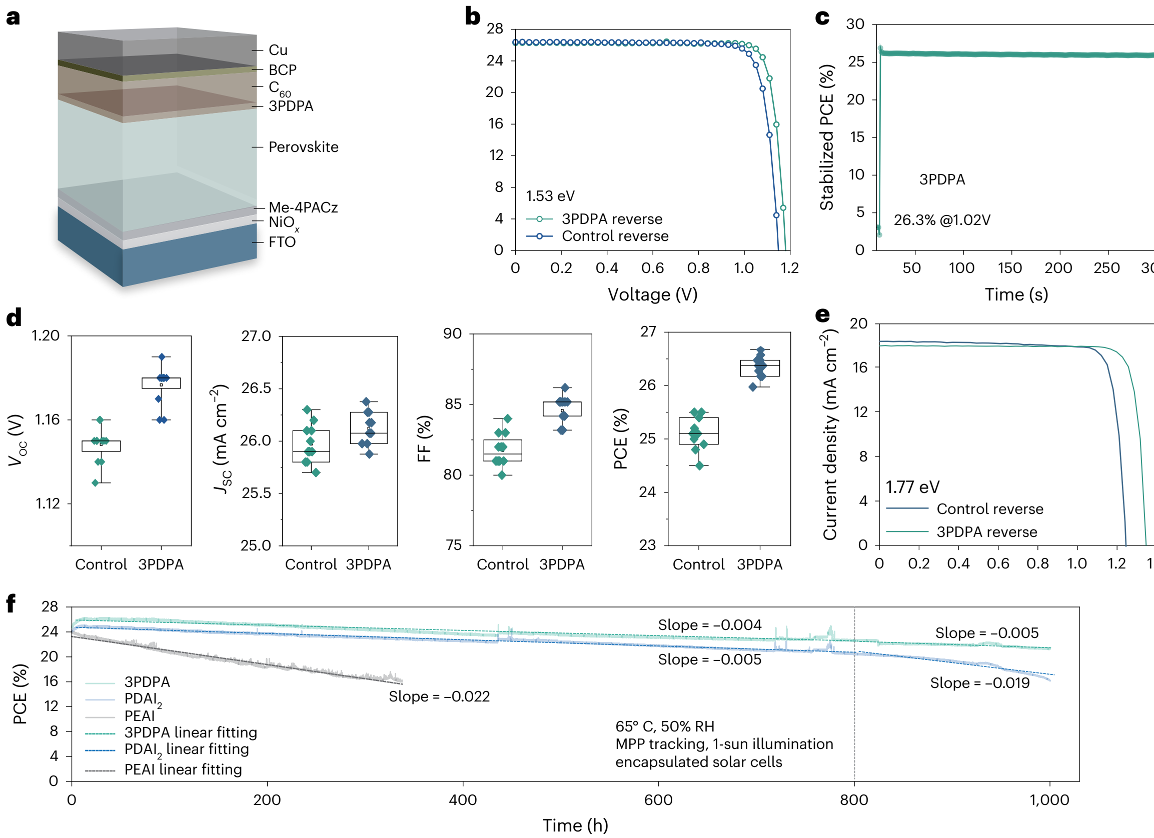

Fig. 4 | Photovoltaic performance and long-term operational stability. **a**, Schematic illustration of the device architecture. **b**, *J–V* curves of the champion 1.53 eV bandgap devices with different interfacial structures in reverse scan mode. **c**, Stabilized PCE of the champion 3PDPA-modified device. **d**, Statistical analysis of key photovoltaic parameters based on 11 cells for each type. **e**, *J–V* curves of 1.77 eV bandgap devices in reverse scan mode. **f**, MPP tracking stability of PSCs with different interlayers under continuous illumination at 65 °C in ambient air (~50% relative humidity). The illumination is provided by a large-area LED array with the intensity calibrated to 1-sun equivalent intensity. The initial PCEs of the PEAI-, $PDAI_2$- and 3PDPA-based devices are 24.0%, 24.2% and 24.7%, respectively.

of 2.5 D (Supplementary Fig. 22), adopts a vertical orientation on both PbI_2^- and FAI^- terminated perovskite surfaces (Supplementary Fig. 23). Its P = O moiety coordinates with undercoordinated Pb^{2+} ions, forming stable Pb–O bonds (bond length 2.4 Å), whereas the aromatic core engages in π–π stacking with C_{60}. Polarization-modulated infrared reflectance absorption spectroscopy confirms this orientation: new absorption bands at 1,410 and 1,770 cm^{-1}, assigned to the –C–N and –C = O groups of 3PDPA, appear only after molecular assembly (Supplementary Fig. 24), consistent with a non-zero normal dipole component. Given the near-coplanarity of the benzamide units in 3PDPA, this observation supports a tilted vertical coverage pattern of 3PDPA on the perovskite surface. This dipole alignment establishes an interfacial energy cascade that reduces the offset between perovskite and C_{60}, enabling energetically favourable yet slowed electron extraction (Supplementary Fig. 15b). Dual-mode coordination and π–π interaction together produce a chemically and electronically coherent interface with an enhanced binding energy of 1.78 eV, confirming the robustness and functionality of the 3PDPA-mediated contact.

Charge density difference plots along the out-of-plane (*z*) axis (Fig. 4g and Supplementary Fig. 25) reveal the spatial characteristics of interfacial charge redistribution. At pristine perovskite/C_{60} interface, charge accumulation is confined to the direct contact region. Upon 3PDPA functionalization, the interfacial region between 3PDPA and perovskite exhibits markedly intensified and spatially extended charge accumulation[48], reflecting strong interactions, consistent with the observed coordination and hydrogen-bonding effects. These interactions localize electrons at the interface, creating a confined interfacial electron cloud that slows the electron extraction rate. Consequently, the carrier density at the perovskite surface is elevated.

Photovoltaic performance

Building on the observed enhancement in QFLS, we now evaluate how 3PDPA improves the photovoltaic performance of PSCs. Devices are fabricated in an inverted configuration with the structure FTO/NiO_x/Me-4PACz/$Cs_{0.05}FA_{0.9}MA_{0.05}PbI_3$/(with or without 3PDPA)/C_{60}/BCP/Cu, where FTO is fluorine-doped tin oxide, NiO_x is nickel oxide and Me-4PACz is [4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Fig. 4a, Supplementary Fig. 26 and Supplementary Table 2). More details are provided in the Supplementary Information. The best-performing control device achieves a power conversion efficiency (PCE) of 25.38% with a short-circuit current density (J_{sc}) of 26.29 mA cm^{-2}, a V_{oc} of 1.15 V and a fill factor (FF) of 84.2%. The champion 3PDPA-based device achieves a current density–voltage (*J–V*) scanned PCE of 26.82%, (26.36% independently verified; Supplementary Fig. 27), with an V_{oc} of 1.18 V, J_{sc} of 26.32 mA cm^{-2} and FF of 86.5% (Fig. 4b and Supplementary Table 2). Under maximum power point (MPP) tracking, the stabilized efficiency reaches 26.3% (Fig. 4c). The external quantum efficiency (EQE) spectrum is presented in Supplementary Fig. 28. Compared to control devices, 3PDPA-modified cells show clear improvements in both V_{oc} and FF, attributed to suppressed interfacial non-radiative recombination and enhanced QFLS. These enhancements are statistically reproduced across multiple devices (Fig. 4d), confirming the reliability of 3PDPA-induced interfacial regulation. In wide-bandgap (1.77 eV) cells, 3PDPA enables a *J–V* scanned PCE of 21.2% (20.75% independently verified; Fig. 4e and Supplementary Fig. 29), with an V_{oc} of 1.34 V, J_{sc} of 17.88 mA cm^{-2} and FF of 88.4%. These results underscore the universality and versatility of 3PDPA across diverse device platforms.

To benchmark the performance of 3PDPA, we compare it with the widely used passivation molecule PEAI[34,41,49–52], which shares π-conjugated headgroup–alkyl-spacer–anchoring-unit architecture (Supplementary Fig. 30). This structural resemblance enables a mechanistic comparison of how the phosphonic acid anchor and benzoyl-imide terminal group influence stability. We evaluate their long-term device stability under continuous 1-sun equivalent illumination at 65 °C in ambient air at ~50% relative humidity, corresponding to ISOS-L-3 conditions (Fig. 4f). We determine that for 3PDPA-modified devices, the most stable device exhibited a 10% degradation lifetime (T_{90}) of 1,000 h. Combined analyses from double-ion injection, TRPL and KPFM measurements reveal that the poor long-term stability of PEAI arises from mobile iodide ions, passivation dysfunction, which readily migrate under thermal or electrical stress[22,53] (Supplementary Figs. 31–33). DFT calculations further reveal that 3PDPA exhibits a substantially higher interfacial binding energy than PEAI (Supplementary Fig. 34). This strong chemical anchoring not only stabilizes the interface against ion migration and thermal perturbation but also underpins the long-term operational durability observed in 3PDPA-modified devices.

We further benchmark 3PDPA against the advanced ammonium-halide passivator $PDAI_2$, under identical ageing conditions. Whereas the encapsulated devices exhibit comparable T_{90} values (Fig. 4f), analysis of degradation kinetics via linear fitting of PCE evolution over time shows distinct behaviours: $PDAI_2$ maintains early-stage stability similar to 3PDPA (slopes: −0.005 and −0.004 over ~800 h) but accelerates thereafter (slope: −0.019, probably due to iodide-induced ion migration[22]), whereas 3PDPA sustains a consistent low degradation rate (~−0.005 throughout). This superiority is amplified in unencapsulated devices under the same stringent conditions (Supplementary Fig. 35), where $PDAI_2$ degrades rapidly (slope: −0.177) while 3PDPA shows minimal linear decay (slope: −0.008), highlighting the enhanced intrinsic interface robustness of 3PDPA and validating its advancement over conventional ammonium-halide passivators for inverted PSCs.

Conclusion

In conclusion, we present a rational interface engineering strategy that redefines electron extraction in perovskite solar cells. The 3PDPA molecule enhances QFLS by enabling a slowed charge extraction—preserving interfacial carrier density while suppressing non-radiative recombination. In addition, it mitigates interfacial potential fluctuations, promotes the ordered growth of C_{60} and reinforces interfacial binding through strong dual-mode interactions. This work complements, rather than contradicts, prior strategies that enhance charge extraction to suppress interfacial recombination and highlights that for well-passivated fullerene-based contacts, tuning the extraction rate provides an additional degree of freedom to optimize QFLS and photovoltage. Together, these effects yield an efficiency of 26.82% with excellent operational stability, addressing long-standing limitations of conventional ammonium-halide passivation.

Methods

Materials

All reagents were purchased from commercial suppliers and used as received without further purification. 3PDPA synthesis reagents were sourced from Bide Pharmatech. Nickel (II) acetylacetonate (Ni(acac)$_2$, 95%) was purchased from Sigma-Aldrich. Formamidinium iodide (FAI, 99.99%) and methylammonium iodide (MAI, 99.99%) were purchased from Greatcell Solar Materials. Caesium iodide (CsI, 99.9%) was purchased from Alfa-Aesar. Methylamine hydrochloride (MACl, 98.0%) was purchased from Advanced Election Technology. 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP, 98%) was purchased from OLED. Lead iodide (PbI_2, 98%) and [4-(3,6-Dimethyl-9H-carbazol-9-yl)butyl] phosphonic acid (Me-4PACz, 99%) purchased from Tokyo Chemical Industry. Buckminsterfullerene (C_{60}, 99.5%) was purchased from Nano-C. Hydrochloric acid (HCl; 36% ~ 38%) was purchased from Sinopharm. The fluorine-doped tin oxide (FTO) and indium-doped tin oxide (FTO) coated glasses were purchased from SUNYO. *N*,*N*-dimethylformamide (DMF, 99.8%) and dimethyl sulfoxide (DMSO, 99.7%) were purchased from Sigma-Aldrich. Anhydrous chlorobenzene (CB, 99.8%), anhydrous ethanol and isopropanol were purchased from Across Organic.

Perovskite solar cell fabrication

1.53-eV bandgap perovskite cell fabrication. FTO substrates (7 Ω per square) were sequentially cleaned by sonication with detergent, deionized water, acetone and isopropyl alcohol for 10 min, respectively. Then the FTO substrates were dried using dry air and subsequently treated with ultraviolet-ozone for 20 min. For the NiO_x films, we used the same deposition method as described in the previous work[54]. After annealing, the FTO/NiO_x substrates were transferred to a nitrogen-filled glovebox, and Me-4PACz (1 mg ml^{-1} in ethanol) was spin coated at 3,000 rpm for 30 s, followed by annealing at 100 °C for 10 min. PbI_2 (1.5 mmol), FAI (1.275 mmol), MAI (0.15 mmol) and CsI (0.15 mM) were mixed corresponding to the chemical formula of $Cs_{0.05}MA_{0.05}FA_{0.9}PbI_3$ with 15% MACl addition in 745-µl DMF and 155-µl DMSO solvent to prepare perovskite precursor solution. The precursor solution was stirred at 60 °C for 1 h and then filtered through a 0.22-µm polytetrafluoroethylene membrane. The perovskite precursor was spin coated at 1,000 rpm for 10 s and 4,500 rpm for 40 s, and 150 µl of CB was dripped 5 s before the end, and then the substrate was annealed at 100 °C for 30 min. 3PDPA (1 mg ml^{-1}) were dissolved in a mixed solvent of IPA and DMF with a volume ratio of 200:1 and then spin coated at 4,000 rpm for 30 s, followed by annealing at 100 °C for 5 min. Finally, C_{60} (20 nm), BCP (7 nm) and copper electrode (100 nm) were deposited sequentially using thermal evaporation under a high vacuum (≤ 8 × 10^{-5} Pa). The chamber temperature was controlled under 25 °C during the whole evaporation process. A ~ 100 nm layer of MgF_2 was also evaporated at the glass side to minimize the reflection losses from the glass substrates.

1.77-eV bandgap perovskite cell fabrication. The $FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})_3$ perovskite precursor solution (1.3 M) was prepared in mixed solvent of DMF/DMSO mixed solvent with a volume ratio of 3:1. One mol% $Pb(SCN)_2$ (relative to Pb) was added into the solution. The perovskite precursor solution was stirred for ~12 h before use. The ITO substrates (10 Ω per square) were cleaned following the same procedure used for the FTO substrates. The substrates were then transferred into a glovebox. 4PADCB (0.8 mg ml^{-1} in ethanol) was spin coated at 4,000 rpm for 40 s and annealed at 100 °C for 10 min. The perovskite precursor solution was spin coated at 500 rpm for 2 s and 4,000 rpm for 60 s with 700 µl diethyl ether dripped at the 25 s in the second spinning step. The as-deposited films were then annealed at 65 °C for 2 min and 100 °C for 10 min. 3PDPA solution (1 mg ml^{-1} in IPA and DMF with a volume ratio of 200:1) was spin coated on the perovskite at 4,000 rpm for 60 s and then annealed at 100 °C for 5 min. Finally, the substrates were transferred to the evaporation chamber. Twenty nm C_{60}, 7 nm BCP and 100 nm Ag were subsequently deposited on top by thermal evaporation.

Device characterizations

The *J*–*V* curves were measured by a 2400 Series Source Meter (Keithley Instruments) in the air under simulated AM 1.5 G sunlight at 100 mW cm^{-2} irradiation. The light source is a AAA-class Xenon lamp solar simulator (WACOM; WXS-100S-5), and the intensity was calibrated using a Fraunhofer ISE-calibrated silicon reference cell with KG1 filter. The mismatch factor was calculated to be less than 1% and applied to calibrate the AM 1.5 G 100 mW cm^{-2} equivalent irradiance. The active area of the solar cells was 0.04 cm^2 for the 1.53 eV cells and 0.07 cm^2 for the 1.77 eV cells, defined by an opaque aperture mask. The J–V curves were measured at a scan rate of 300 mV s^{-1} (with no preconditioning) in ambient air (~25 °C and relative humidity ~50%). To achieve the reported peak PCE, devices were subjected to 30 min of light soaking, followed by storage in a desiccator at relative humidity below 30% for 1 ~ 2 days. The external quantum efficiency (EQE) was measured in ambient air using the Zolix-SSC600 system, calibrated by a Si reference cell.

Stability testing
The devices were sealed in a nitrogen glovebox using cover glass and ultraviolet curing adhesive (Lumec LT-U001) and placed in a third-party stability system for stability testing. For long-term ageing tests, a UV-containing large-area light-emitting diode array was used as the light source. The light intensity was calibrated to match 1-sun conditions. The spectrum of the LED simulator used in maximum power point tracking (MPPT) stability test is shown in our previous work[54]. The devices were placed in a fixture and heated using a hot bench to maintain devices at 65 °C in ambient air with a relative humidity of ~50%. We note that the tested devices were encapsulated in a nitrogen-filled glovebox without prior light soaking. To minimize potential degradation from lateral moisture or oxygen ingress, the device edges were removed before electrode deposition, allowing the UV adhesive to contact the FTO substrate directly. For MPP tracking, as V_{MPP} may vary over time, we applied perturbation and observation algorithms to periodically perturb the operating voltage to maximize the total power output of the device.

Other characterizations
Time-resolved photoluminescence (TRPL) spectra were recorded with a time-correlated single-photon counter (PicoQuant GmbH, TimeHarp 260), excited with a picosecond diode laser (640 nm, LDH-P-C-640B) at a repetition rate of 125 kHz. The laser excitation fluence is 3.2 nJ cm^{-2}. The ^1H and ^{13}C NMR spectra. ^1H NMR and ^{13}C NMR were conducted in DMSO-d_6/$CDCl_3$-d using a Bruker AVANCE NEO 600 instrument. High-resolution solid-state nuclear magnetic resonance (ss-NMR) spectroscopy was carried out on a Bruker Avance Neo 400 MHz spectrometer with a 4.0 mm double-resonance MAS probe under a spinning speed of 10 kHz. Single Crystal X-ray Diffractometer measurements were conducted on a XtaLAB Synergy Custom and visualized used Mercury software[55]. X-ray Photoelectron Spectroscopy (XPS) measurements were carried out on a Thermo Fisher ESCALAB 250Xi instrument utilizing a monochromatic Mg K Alpha (1.254 keV) X-ray source, with samples measured under ultra-high vacuum conditions. Binding energies were calibrated using the C (1s) carbon peak at 284.8 eV. Fitting was performed by Avantage. Fourier transform infrared (FTIR) spectroscopy of the samples were characterized using Thermo Fisher Nicolet iS10. Double-ion injection (DI) current measurements on the devices were biased with a 1.5 V modulated forward bias voltage generated by an arbitrary function generator (Agilent, 33612 A). The repetition rate used during testing was 50 Hz. TRPL spectra were recorded using a time-dependent single-photon counter (PicoQuant GmbH, TimeHarp 260) and excited at a repetition rate of 125 kHz using a picosecond diode laser (640 nm, LDH-P-C-640B). PLQY values were recorded by a commercialized system (XPQY-EQE, Guangzhou Xi Pu Optoelectronics Technology Co. Ltd.). The atomic force microscope was tested by Park NX 10. Polarization-resolved grazing incidence infrared reflection spectroscopy was performed on a Fourier transform infrared spectrometer (INVENIO R, Bruker) equipped with photoelastic modulator (PEM-100, Hinds Instruments) and liquid-nitrogen-cooled mercury cadmium telluride detector. The incidence angle was set to 85°, with a spectral resolution of 4 cm^{-1}. Each final spectrum was obtained by averaging 64 scans. The work function was measured using a Kelvin probe system (KP Technology, KP020), highly oriented pyrolytic graphite flakes were used to calibrate the probe tip, and the Fermi level is subsequently determined based on the calibrated measurements. The transient absorption (TA) spectroscopy was performed by a pump-probe spectrometer (HARPIA Light Conversion). Fundamental laser with a centre wavelength of 1,030 nm and pulse duration ~ 260 fs was generated at a repetition rate of 1 kHz (PHAROS-PH2, Light Conversion). Then the fundamental laser was split into two parts, one part is guided into an optical parameter amplifier (ORPHEUS-HP, Light Conversion) to create a pump laser from 315 nm to 2,600 nm. The frequency of the pump laser was chopped at 100 Hz. The other part of the laser was focused into a sapphire crystal to generate 450–1,100 nm probe laser, respectively. The samples excited by the pump pulse were also sampled by the probe. The time delay (t) between the pump and probe was systematically varied to monitor the evolution of the absorption profile over time caused by the pump. A difference absorption spectrum (ΔA) was calculated by,

$$\Delta A = \log \frac{P_0}{P(t)},$$

where P_0 and $P(t)$ are pre-pump power and post-pump power, respectively. By changing the t between the pump and the probe and recording a ΔA spectrum at each time delay, a ΔA profile as a function of the t and wavelength (λ) was obtained. It is noted that the temporal overlap between pump and probe was wavelength dependent, due to the chiro of white light. All 2D TA data have been corrected by cross correlation of the continuum with the pump pulse.

Device simulation and computing
Numerical simulations were performed using the solar cell capacitance simulator in one dimension (SCAPS-1D) to model a complete FTO/HTL/Perovskite/C_{60}/Cu device structure. To investigate the effect of carrier density near the perovskite/C_{60} interface, the perovskite layer was divided into two sublayers and a 30-nm-thick interfacial region adjacent to the C_{60} contact was introduced. The doping concentration of this interfacial region was varied logarithmically from 1×10^{16} to 1×10^{18} cm^{-3} across 20 points to simulate the carrier accumulation resulting from different electron extraction scenarios. In parallel, the interfacial recombination velocity at the perovskite/C_{60} contact was swept logarithmically from 10^0 cm s^{-1} to 10^3 cm s^{-1} using four points to represent different levels of interfacial defect densities. For each condition, the open-circuit voltage (V_{oc}), carrier density (n) and quasi-Fermi levels (E_{Fn}, E_{Fp}) within the device were recorded. These datasets were used to construct the V_{oc} – n and E_F splitting plots in Fig. 3d,e, respectively. The first-principles calculations about the differential charge density and binding energy were performed using the projector augmented wave method[56] implemented in the Vienna Ab initio Simulation Package code[57]. The electron exchange-correlation energy was addressed employing the generalized gradient approximation as formulated by Perdew, Burke and Erzerhof[58]. $FAPbI_3$ in the model was constructed by (2 × 2 × 3) unit cells for the interaction with 3PDPA and C_{60}. The Monkhorst–Pack scheme k-point grid of (2 × 2 × 1) was used for relaxation. Plane-wave basis with a cut-off energy of 600 eV was adopted and the convergence of the total ground-state energy was achieved within a threshold of 10^{-6} eV. The optimal model was employed for the electronic structure to further clarify the intrinsic properties of the materials. All calculations are visualized using VESTA software[59].

Reporting summary
Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability
The data that support the findings of this study are available within the Article and its Supplementary Information. Source data are provided with this paper.

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Author contributions

The project was under the supervision of Z.W. M.L. and Z.W. and conceived the project and designed most of the experiments. M.L. synthesized the molecule. Y.Y. and M.L. performed the device preparation. S.L. prepared wide-bandgap devices. X.Y. performed DFT calculations under the supervision of S.Y. Y.Z. assisted in the synthesis of the molecule. S.C. performed the simulations with assistance of J.F. and contributed to the data analysis. H.L. assisted in testing the PLQY. Q.Y. assisted in testing KPFM. Z.X. tested the atomic force microscope. H.F. assisted in testing the polarized infrared spectroscopy under the supervision of C.G. Y.L. performed XPS measurement. Q.L. provided TA testing and analysed data. M.L. and Z.W. wrote the first draft of the paper. All the authors contributed to the revision of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Solar Cells Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form is intended for publication with all accepted papers reporting the characterization of photovoltaic devices and provides structure for consistency and transparency in reporting. Some list items might not apply to an individual manuscript, but all fields must be completed for clarity.

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Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells
- [x] Yes
- [ ] No

The area of the tested cells is 0.04 cm^2 for the 1.53 eV cells and 0.07 cm^2 for the 1.77 eV cells.

Explain why this information is not reported/not relevant.

Method used to determine the device area
- [x] Yes
- [ ] No

The device area is defined by an aperture mask.

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction
- [x] Yes
- [ ] No

Provided in this Fig. 4b and Supplementary Table 2.

Voltage scan conditions
- [x] Yes
- [ ] No

The J–V curves were measured at a scan rate of 300 mV/s.

Explain why this information is not reported/not relevant.

Test environment
- [x] Yes
- [ ] No

The cell are tested in air (~25 °C and relative humidity ~50%).

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization
- [x] Yes
- [ ] No

No preconditioning.

Explain why this information is not reported/not relevant.

Stability of the J-V characteristic
- [x] Yes
- [ ] No

Stabilised PCE was provided in Fig. 4c.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization
- [x] Yes
- [ ] No

Very minor hysteresis was observed for devices herein.

Explain why this information is not reported/not relevant.

Related experimental data
- [x] Yes
- [ ] No

J-V curves under reverse and forward scans were provided in supplementary Table 2.

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)
- [x] Yes
- [ ] No

EQE curves were provided in Supplementary Fig. 28.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator
- [x] Yes
- [ ] No

The integrated Jsc values from EQE differ by less than 5% from those measured by J-V scans, which is within the accuracy confidence of the measurements.

Explain why this information is not reported/not relevant.

For tandem solar cells, the bias illumination and bias voltage used for each subcell

☐ Yes — Provide a description of the measurement conditions.

☒ No — Not relevant.

5. Calibration

Light source and reference cell or sensor used for the characterization

☒ Yes — Described in Method section.

☐ No — Explain why this information is not reported/not relevant.

Confirmation that the reference cell was calibrated and certified

☒ Yes — The reference cells were calibrated by Fraunhofer ISE, as described in Method.

☐ No — Explain why this information is not reported/not relevant.

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes — The mismatch factor was calculated to be less than 1% and applied to calibrate the AM 1.5G 100mW cm^−2 equivalent irradiance.

☐ No — Explain why this information is not reported/not relevant.

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes — Metal Aperture areas of 0.04 cm^2 for 1.53 eV perovskite and 0.07 cm^2 for 1.77 eV perovskite were used in this work.

☐ No — Explain why this information is not reported/not relevant.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes — Report the difference in the short-circuit current density values measured with the mask and aperture area.

☒ No — All J-V curves were measured with a metal aperture mask.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes — CCIC Southern Testing Co. Ltd. and Chengdu Institute of Product Quality Inspection Co. Ltd.

☐ No — Explain why this information is not reported/not relevant.

A copy of any certificate(s)

☒ Yes — Supplementary Figures 27 and 29.

☐ No — Explain why this information is not reported/not relevant.

8. Statistics

Number of solar cells tested

☒ Yes — 11 cells per type.

☐ No — Explain why this information is not reported/not relevant.

Statistical analysis of the device performance

☒ Yes — Fig. 4d.

☐ No — Explain why this information is not reported/not relevant.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

☒ Yes — Long-term device stability of encapsulated cells were measured under continuous 1-sun equivalent illumination at 65 °C in ambient air with a relative humidity of ~50% (described in Fig. 4f caption and Method section).

☐ No — Explain why this information is not reported/not relevant.